Credit Risk Analysis — Business Insights

Project-based analysis of customer default behavior, risk segments, and exposure at risk

Source: User-provided banking credit-risk project notebook and the outputs supplied in the conversation. No machine-learning model is used in this report.

Executive Summary

The dataset contains 10,000 customers, of which 1,500 are observed as defaulted and 8,500 are non-defaulted. The overall observed customer default rate is **15.00%**. Total outstanding exposure is ■**484.45 crore**, of which ■**68.10 crore** is associated with defaulted customers, giving an exposure-at-risk share of **14.06%** under the project's definition.

The strongest observed differences are lower credit score and much longer delinquency among defaulted customers. Defaulted customers have an average credit score of 680.09 versus 714.26 for non-defaulted customers, and average delinquency of 75.47 days versus 15.61 days. DTI and utilization are almost identical between the two groups, so the supplied aggregate output does not support treating them as strong differentiators.

Interpretation: “Likely to default” below means customers with characteristics associated with observed defaults in this dataset. It is not a future-default prediction because no ML model or out-of-sample probability was used.

1. Who is likely to default?

Based on the observed data, the defaulted population is characterized by lower credit scores and substantially higher delinquency. The clearest customer profile is therefore a customer with a weaker credit-score band and/or prolonged delinquency. Repeated missed payments become particularly notable at 3–4 missed payments, where the observed default rate is 27.00%.

Metric	Non-defaulted (0)	Defaulted (1)	Difference
Customers	8,500	1,500	—
Average credit score	714.26	680.09	-34.17
Average DTI	38.00%	38.11%	+0.11 pp
Average missed payments	0.70	0.82	+0.12
Average delinquency days	15.61	75.47	+59.86 days
Average utilization	28.49%	28.41%	-0.08 pp
Average outstanding	■10.12 lakh	■9.66 lakh	≈ -■0.46 lakh

> **Insight:** The difference in delinquency is especially large: defaulted customers average about 4.8× the delinquency days of non-defaulted customers. Credit score is also materially lower in the defaulted group. By contrast, average DTI and utilization are nearly the same, so the current output does not show a strong standalone relationship for those two measures.

2. Why is the customer risky?

The supplied customer-level examples show High Risk customers with low credit scores and 90 days of delinquency. For example, the displayed High Risk rows include credit scores from 472 to 565 and 90 delinquency days, with default_flag = 1. This supports the broader aggregate pattern that severe delinquency and weak credit score coincide with observed default.

However, the displayed rows also show that DTI can vary substantially within High Risk customers (for example, values around 2.68% to 70.65%). Therefore, DTI alone does not explain the High Risk classification in the supplied examples.

3. Which customer/loan segments have the highest default rate?

Credit-score segments:

Segment	Customers	Defaults	Default rate
Poor	289	289	100.00%
Fair	2,487	332	13.35%
Good	3,897	480	12.32%
Very Good	2,348	302	12.86%
Exceptional	979	97	9.91%

Missed-payment segments:

Missed payments	Customers	Defaults	Default rate	Defaulted exposure
0	4,859	707	14.55%	■33.39 cr
1	3,529	509	14.42%	■21.48 cr
2	1,269	187	14.74%	■8.51 cr
3–4	337	91	27.00%	■3.78 cr
5+	6	6	100.00%	■0.94 cr

Delinquency segments:

Delinquency	Customers	Defaults	Default rate	Defaulted exposure
1–30 days	3,821	128	3.35%	■7.34 cr
31–60 days	1,232	43	3.49%	■1.84 cr
61–90 days	1,206	1,206	100.00%	■53.30 cr
No delinquency	3,741	123	3.29%	■5.62 cr

Important validation caveat: The 61–90-day delinquency group has 1,206 customers and all 1,206 are defaulted. Likewise, the Poor credit-score group and High Risk band each have 100% observed default. These may reflect how the project's risk/default rules are constructed, so they should not be presented as independent predictive evidence without checking the feature definitions and construction logic.

Customer-segment comparison

Segment	Customers	Defaults	Default rate	Exposure at risk
High Net Worth	426	67	15.73%	13.18%
Premium	1,336	207	15.49%	14.47%
Affluent	2,500	375	15.00%	13.55%
Mass Market	5,738	851	14.83%	14.25%

Customer segment default rates are relatively close (14.83%–15.73%). Therefore, the supplied output does not support saying that one customer segment is dramatically riskier than the others.

4. How much exposure is at risk?

Total outstanding exposure is ■**4,844,545,602.98 (■484.45 crore)**. Defaulted exposure is ■**680,996,263.75 (■68.10 crore)**. Therefore, the project's exposure-at-risk measure is **14.06%**.

Risk band	Customers	Total exposure	Defaulted exposure	Exposure at risk
High Risk	289	■14.06 cr	■14.06 cr	100.00%
Medium Risk	2,487	■121.81 cr	■15.73 cr	12.91%
Low Risk	3,897	■185.04 cr	■18.93 cr	10.23%
Very Low Risk	3,327	■163.54 cr	■19.38 cr	11.85%

A notable portfolio insight is that High Risk is not the only source of exposure. Low Risk and Very Low Risk groups together have ■38.31 crore of defaulted exposure (■18.93 cr + ■19.38 cr), which is about 56.25% of the total defaulted exposure. This means exposure monitoring should consider both default rate and portfolio size.

Definition caveat: ■68.10 crore is defaulted outstanding exposure, not an accounting loss. Actual credit loss would require recovery, collateral, LGD, provisioning, and related information, which is not part of the supplied output.

5. What patterns can help the bank reduce future credit losses?

Early-warning monitoring for delinquency

The 61–90-day delinquency group has 100% observed default, while 1–30, 31–60 and no-delinquency groups are around 3.29%–3.49% in the supplied table. The bank can use increasing delinquency as an early-warning trigger, while checking the dataset definition because the 100% result may be mechanically linked to default.

Monitor repeated missed payments

Customers with 3–4 missed payments have a 27.00% observed default rate versus roughly 14.4%–14.7% for 0–2 missed payments. The 5+ group is 100%, but it contains only 6 customers, so that percentage should not be generalized.

Use credit score together with payment behavior

The Poor credit-score group has 100% observed default and ■14.06 crore exposure. Across the larger bands, default rate declines from 13.35% in Fair to 9.91% in Exceptional. Credit score is therefore useful as a segmentation variable, but should be interpreted with delinquency and payment behavior.

Review large exposures, not only high-risk labels

The 25L+ exposure band contains ■204.30 crore total outstanding and ■26.87 crore defaulted exposure. Its default rate is 13.96%, not the highest rate, but its absolute defaulted exposure is large. Portfolio monitoring should therefore combine default rate with exposure concentration.

Do not over-focus on customer segment

High Net Worth, Premium, Affluent and Mass Market have observed default rates from 14.83% to 15.73%. The relatively narrow range suggests that risk management should prioritize behavioral and credit-risk variables rather than customer segment alone.

Portfolio metrics at a glance

Measure	Verified project output
Customers	10,000
Defaulted customers	1,500
Overall observed default rate	15.00%
Total outstanding exposure	■484.45 crore
Defaulted outstanding exposure	■68.10 crore
Exposure at risk	14.06%
Defaulted average credit score	680.09
Non-defaulted average credit score	714.26
Defaulted average delinquency	75.47 days
Non-defaulted average delinquency	15.61 days

Final Business Conclusion

The project shows an observed default population of 15% and ■68.10 crore of defaulted outstanding exposure against ■484.45 crore total outstanding exposure. The clearest observed risk patterns are lower credit score, prolonged delinquency, and repeated missed payments. The largest absolute exposure risk is not confined to the High Risk band: lower-labelled risk groups also contain substantial defaulted exposure. For future loss reduction, the bank should use delinquency and missed-payment behavior as early-warning signals, combine them with credit-score information, and monitor high-value exposure concentrations. The analysis should be described as historical/observed risk analysis rather than a predictive ML model.

Methodology & Scope

This report uses only the outputs supplied by the user and the attached project notebook. Calculations such as overall default rate, exposure-at-risk percentage, and differences between defaulted/non-defaulted groups are arithmetic checks of those outputs. No external data, machine-learning model, ROC-AUC, probability-of-default model, LGD, or expected-loss model was introduced.